Theorem 6.8.1. *There exists $m \in \mathbb{N}$, such that*

$$d\left(V^{(m)}\right) = \alpha_p$$

for some p.

Proof. If $V^{(i)}$ is surjective at the appropriate vertex k, then

$$d\left(V^{(i+1)}\right) = d\left(F_k^+ V^{(i)}\right) = s_k d\left(V^{(i)}\right).$$

This implies that if $V^{(0)}, \ldots, V^{(i-1)}$ are surjective at the appropriate vertices, then

$$d\left(V^{(i)}\right) = \ldots s_{n-1} s_n d(V).$$

By Lemma 6.7.2 this cannot continue indefinitely, since $d\left(V^{(i)}\right)$ may not have any negative entries. Let i be the smallest number such that $V^{(i)}$ is not surjective at the appropriate vertex. By Proposition 6.6.7 it is indecomposable. So, by Proposition 6.6.5, we get

$$d(V^{(i)}) = \alpha_p$$

for some p. □

We are now able to prove Gabriel's theorem. Namely, we get the following corollaries.

Corollary 6.8.2. *Let Q be a quiver, and let V be any indecomposable representation. Then $d(V)$ is a positive root.*

Proof. By the proof of Theorem 6.8.1

$$s_{i_1} \ldots s_{i_m} (d(V)) = \alpha_p.$$

Since the s_i preserve B, we get

$$B(d(V), d(V)) = B(\alpha_p, \alpha_p) = 2.$$

□

Corollary 6.8.3. *Let V, V' be indecomposable representations of Q such that $d(V) = d(V')$. Then V and V' are isomorphic.*